

Pressure-State relationships for Type 2 data

ALL INDICATORS UPLOADED BEFORE 12 FEBRUARY 2026

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Datasets were downloaded from the ICES-WKBENTH4 Sharepoint on 12 February 2026. Indicators were provided by workshop participants and included: fraclong, invsimpson, longlivedbiomass_to_ref, medlong, rel margalef div (biom), rel margalef div (dens), richness, shannon, simpson, sos, sos_2026, total_abundance, total_biomass, total_biomass_withceph(Table 1).

Type-2 data are pressure-gradient data with minimal environmental variation. Datasets were only analyzed for samples across a fishing pressure gradient ranging from low/none to high fishing intensity. Low-pressure samples were identified using the 15 % quantiles. The threshold for low pressure is set by the data provider and may result in high values. The 15 % quantile was nevertheless considered as reference for low pressure. To enable comparability across indicators and datasets, each indicator value was standardized by dividing its value by the mean value of the indicators for the low-disturbance samples.

The pressure-state relationships were estimated by fitting a log-normal model with the logarithm of each indicator as a linear function of pressure:

$$\log(\text{indicator}) = \beta \cdot \text{pressure} + \varepsilon$$

with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

Pressure-state relationships were tested for 13 indicators. A negative slope implied a decrease in indicator in function of increasing fishing pressure. Indicators with a decreasing state with increasing fishing pressures in at least half of the tested datasets were total_biomass, sos_2026, rel margalef div (dens), longlivedbiomass, invsimpson, medlong, shannon

Table 1: Significant ($P > 0.05$) slopes of the pressure-state relationships for all indicators and datasets. NS = No significant relationship; / = Indicator not estimated.

Dataset ID	Sampling Gear	Total Biomass	SOS2026	Relative Margalef (Density)	Long-lived Biomass	Inverse Simpson	Median Longevity	Shannon	Richness	Long-lived Fraction	Simpson	SOS	Total Abundance	Relative Margalef (Biomass)	N (Indicators)	N (Neg. Slopes)	% (Neg. Slopes)
NS_BIN_beam	trawl	-0.14	/	-0.03	-0.32	/	/	/	-0.05	/	/	/	-0.25	-0.03	6	6	100
BS_gotland	grab	-0.26	-0.14	-0.11	NS	-0.17	-0.13	-0.2	-0.1	-1.03	-0.21	-1.03	NS	-0.08	13	11	85
NS_fiadenground	grab	-0.12	-0.49	-0.08	-2.25	-0.32	-0.12	-0.08	-0.08	-1.47	-0.03	-0.31	NS	NS	13	11	85
NS_BIN_grab	grab	-0.16	/	-0.02	-0.31	/	/	/	-0.04	/	/	/	-0.17	NS	6	5	83
BoBIC_GulfofCadizmud	trawl	NS	/	-0.03	-0.5	/	/	/	-0.02	/	/	/	0.02	-0.03	6	4	67
NS_Kattegat	grab	-0.2	-0.05	0.01	-0.39	/	/	/	NS	/	/	/	-0.04	0.02	7	4	57
NS_Hinderbanken	grab	-0.73	/	-0.37	NS	NS	/	/	NS	/	/	/	-0.23	NS	6	3	50
NS_SilverPit	grab	NS	/	-0.01	-0.08	-0.01	-0.01	-0.01	NS	NS	0	NS	NS	NS	12	5	42
NS_LongForties	grab	-0.87	-8.36	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	13	2	15
BS_southernbaltic	grab	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	13	0	0
BoBIC_GulfofCadizsand	trawl	NS	NS	0.18	NS	/	/	/	0.15	/	/	/	NS	0.2	7	0	0
NS_doggerbank	grab	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	0.16	13	0	0
WMS_DMPEC	grab	/	/	0.11	/	/	/	/	0.14	/	/	/	0.18	/	3	0	0
N(id)		12	7	13	12	6	6	6	13	6	6	6	13	12			
N(neg slope)		7	4	7	6	3	3	3	5	2	2	2	4	3			
%(neg slope)		58	57	54	50	50	50	50	38	33	33	33	31	25			

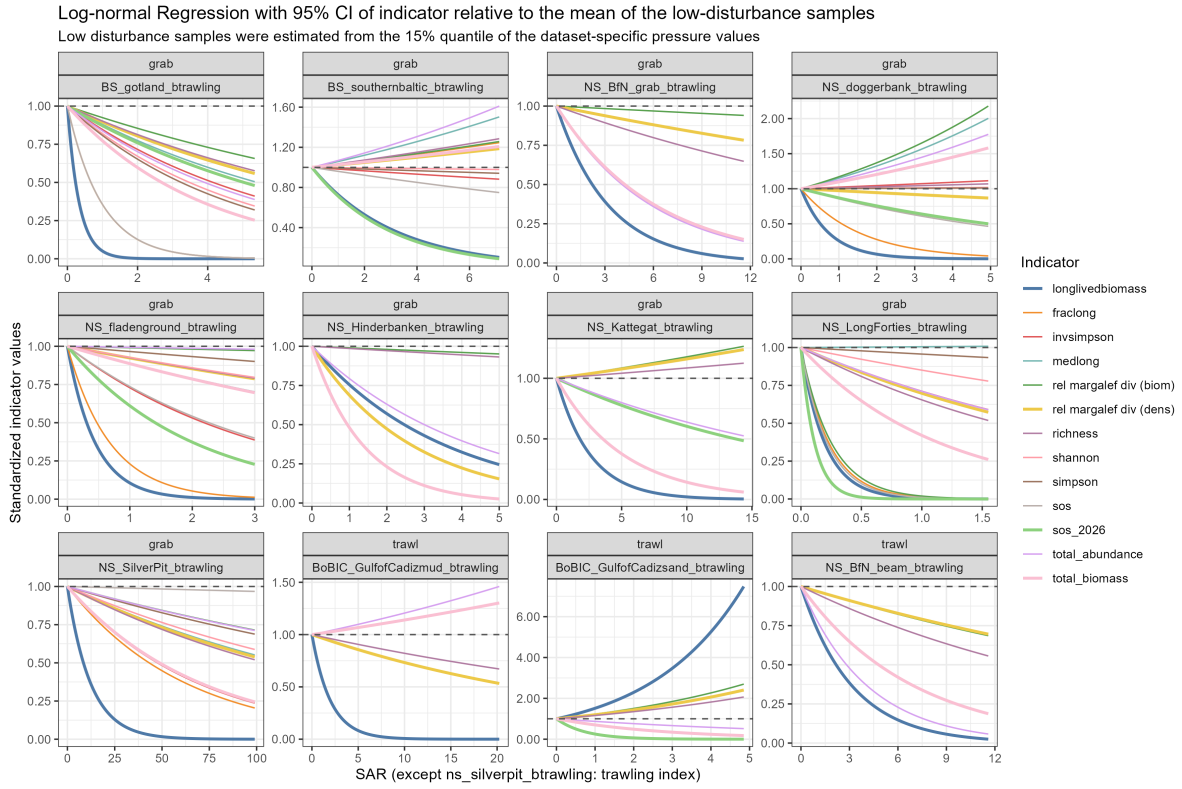


Figure 1: Pressure-state response curves for all indicators.

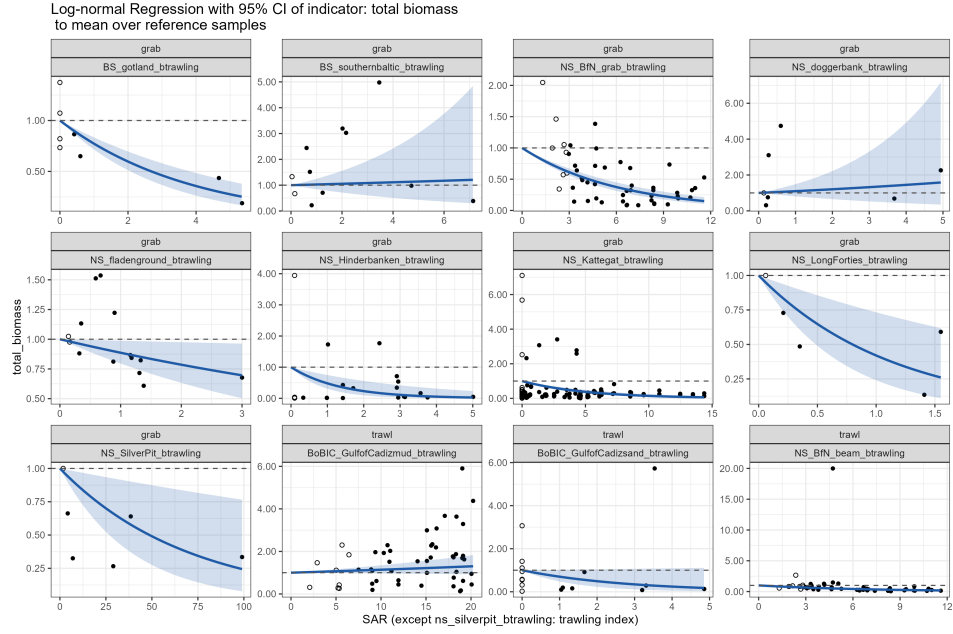


Figure 2: Pressure-state response curves for total biomass.

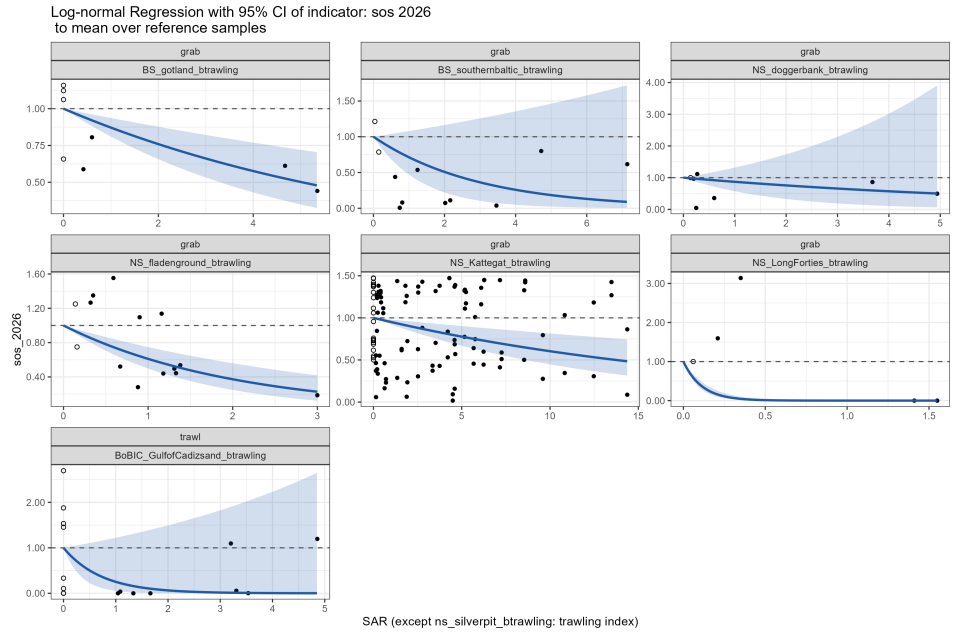


Figure 3: Pressure-state response curves for SOS (2026).

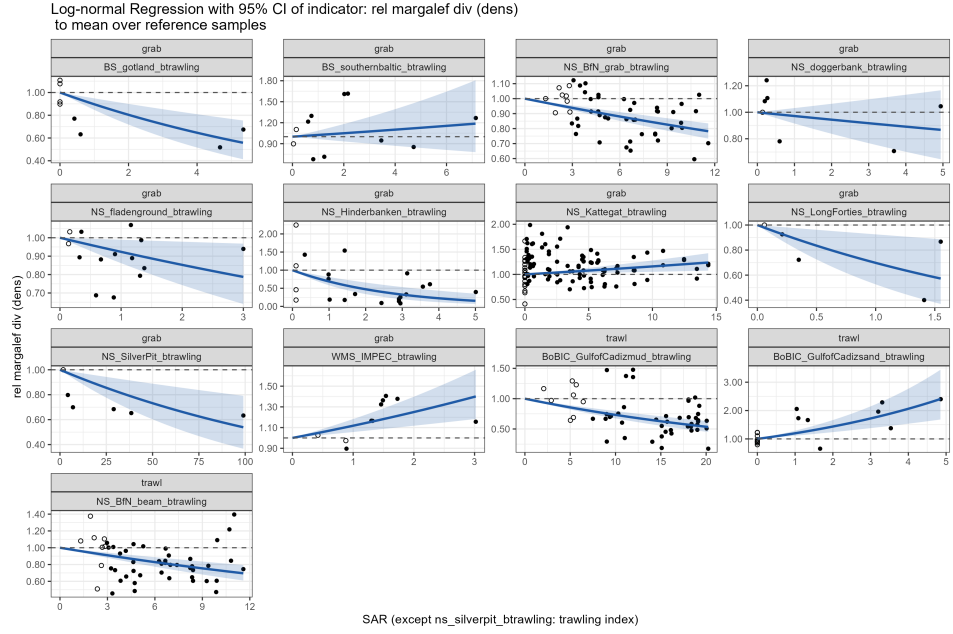


Figure 4: Pressure-state response curves for relative Margaleff diversity (density).

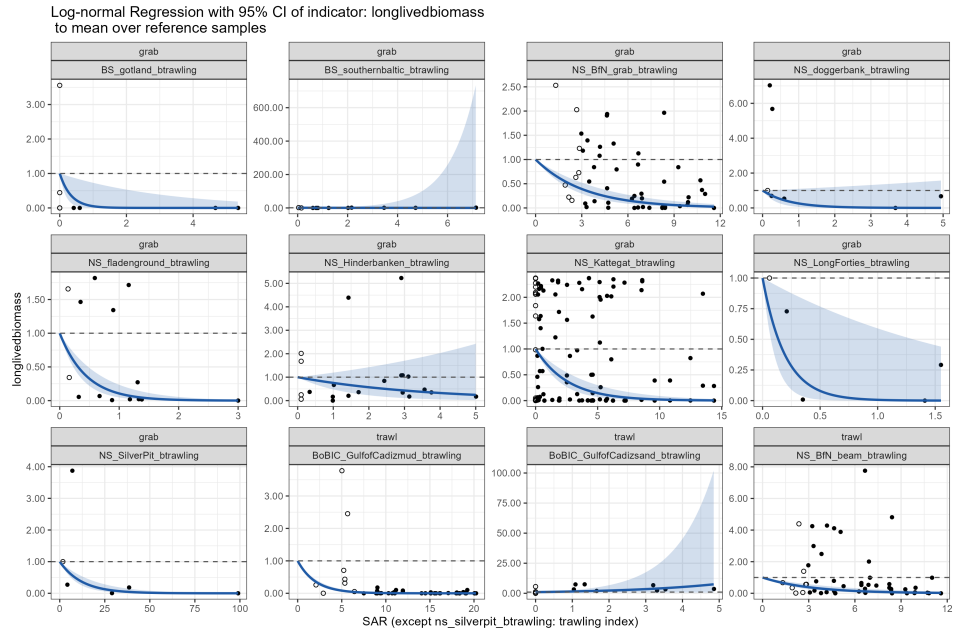


Figure 5: Pressure-state response curves for .

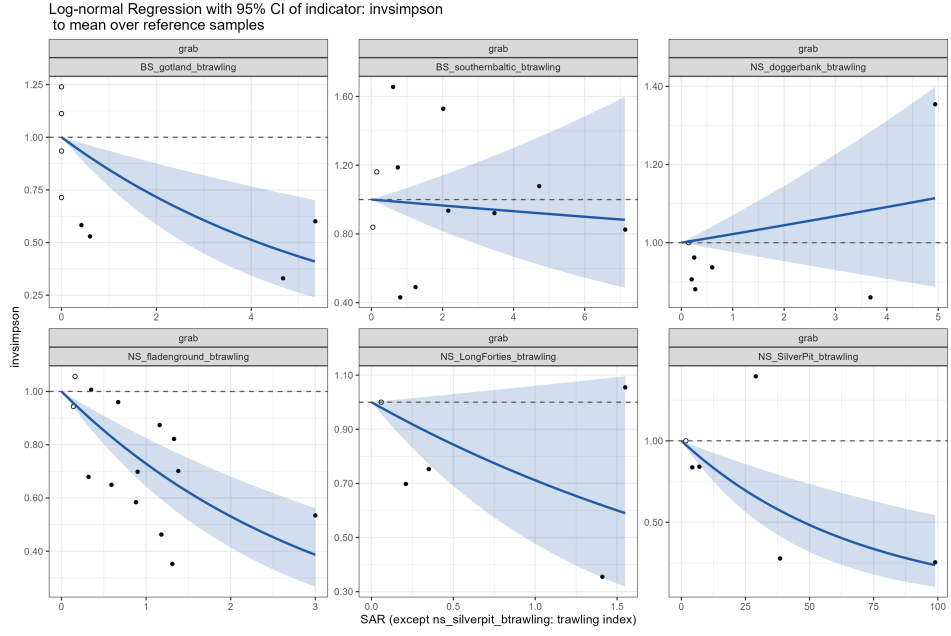


Figure 6: Pressure-state response curves for inverse simpson.

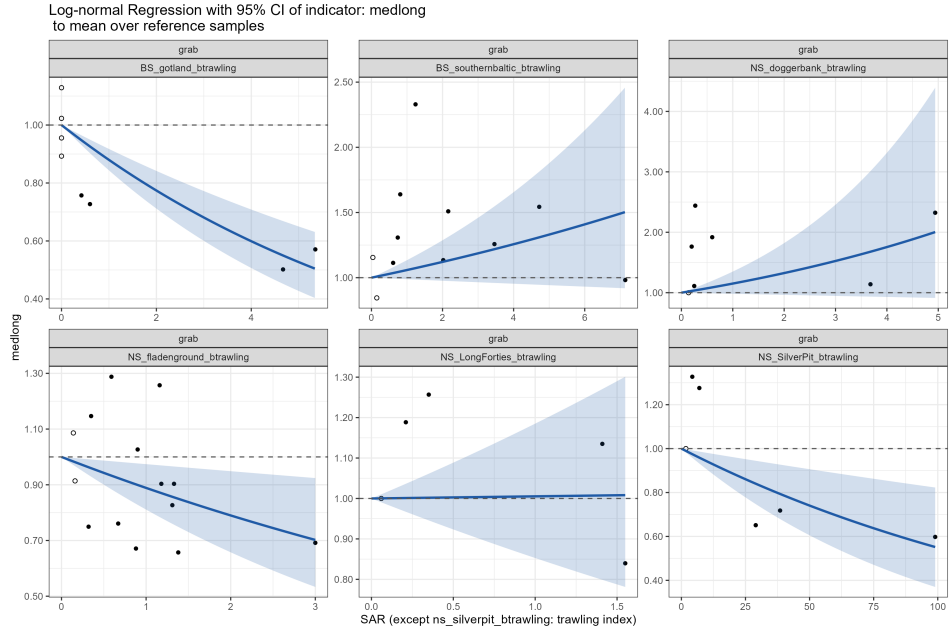


Figure 7: Pressure-state response curves for median longevity.

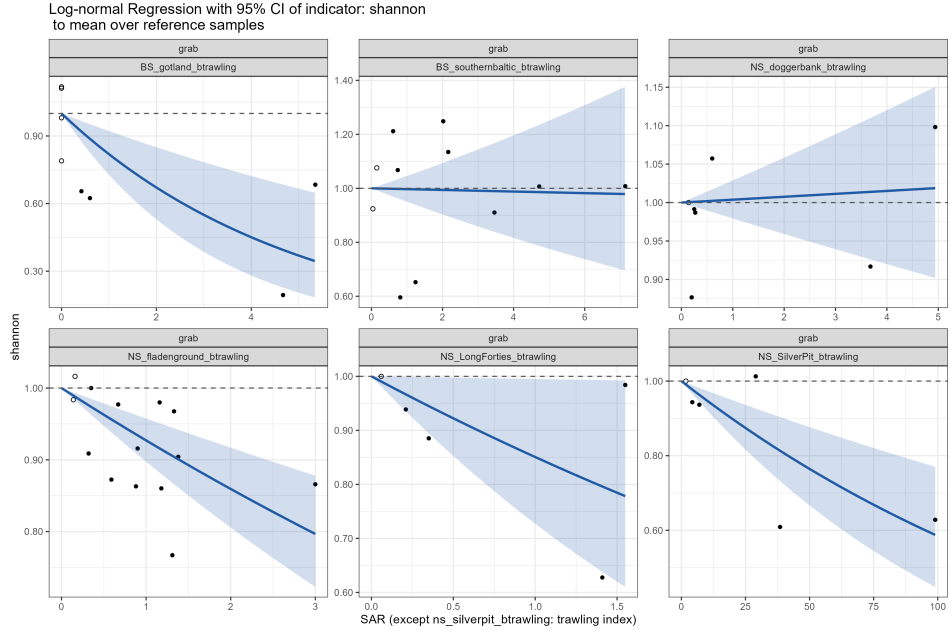


Figure 8: Pressure-state response curves for Shannon diversity.

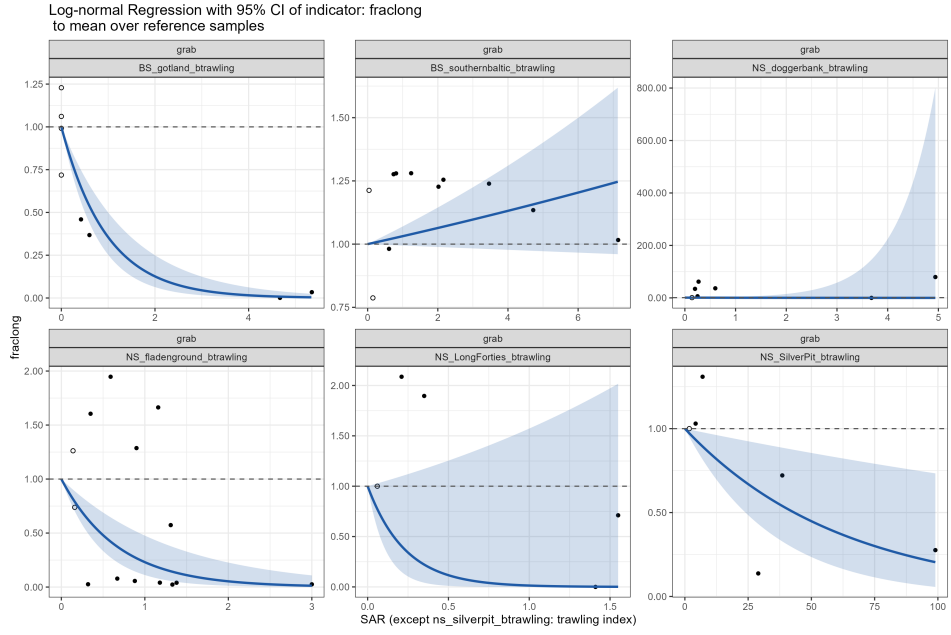


Figure 9: Pressure-state response curves for Fraction long-lived.

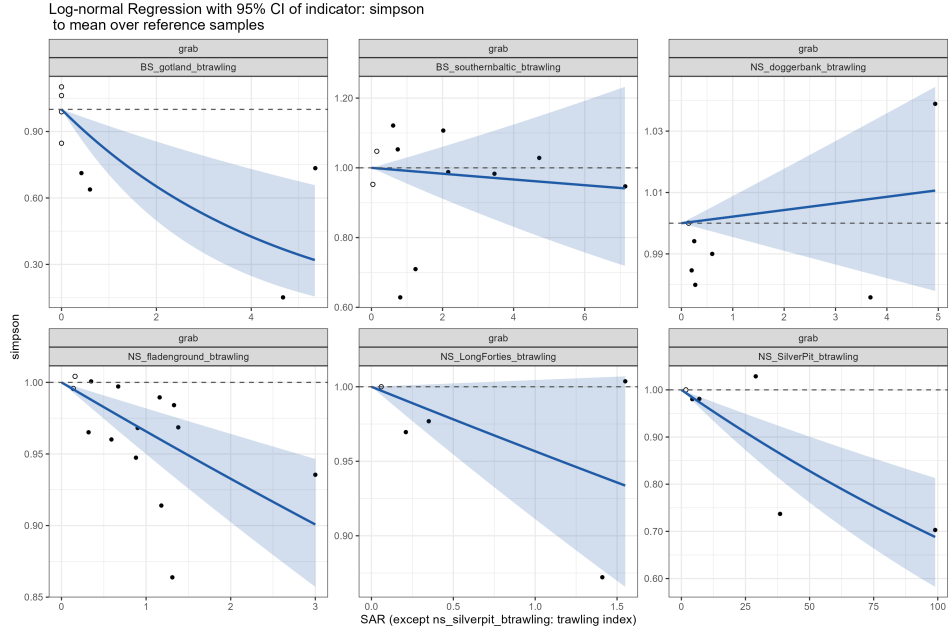


Figure 10: Pressure-state response curves for Simpson diversity.

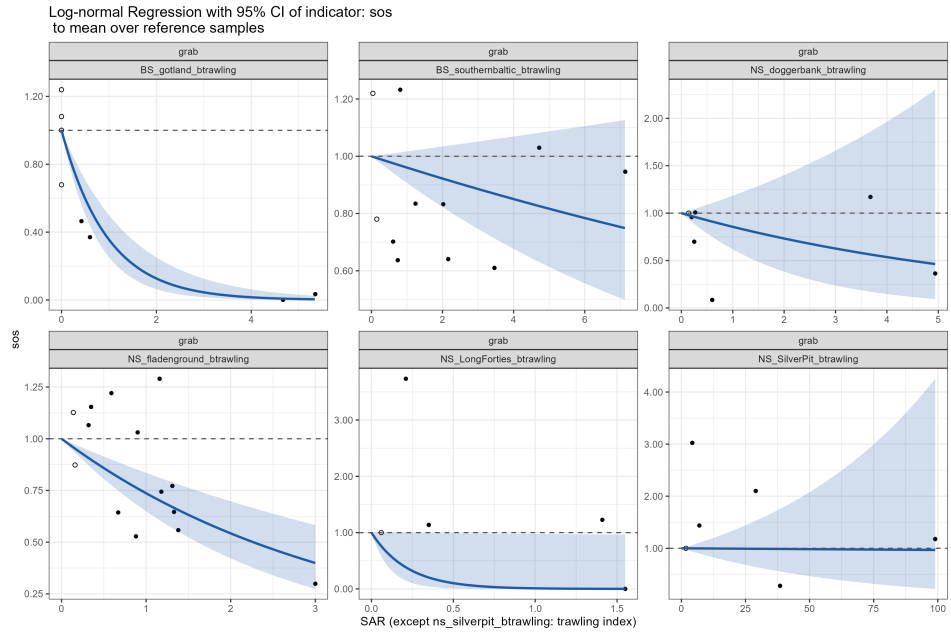


Figure 11: Pressure-state response curves for SOS.

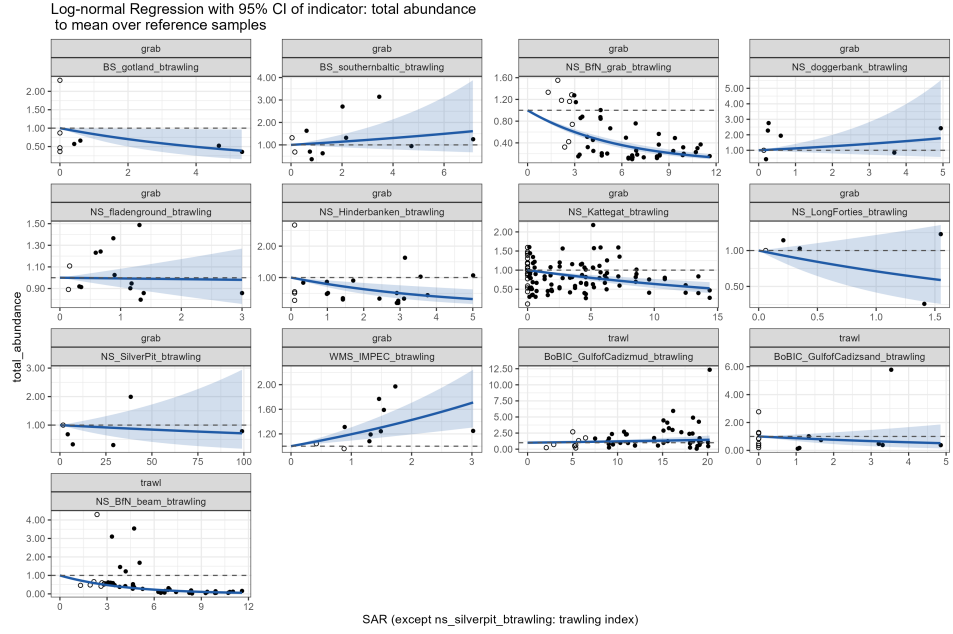


Figure 12: Pressure-state response curves for total abundance.

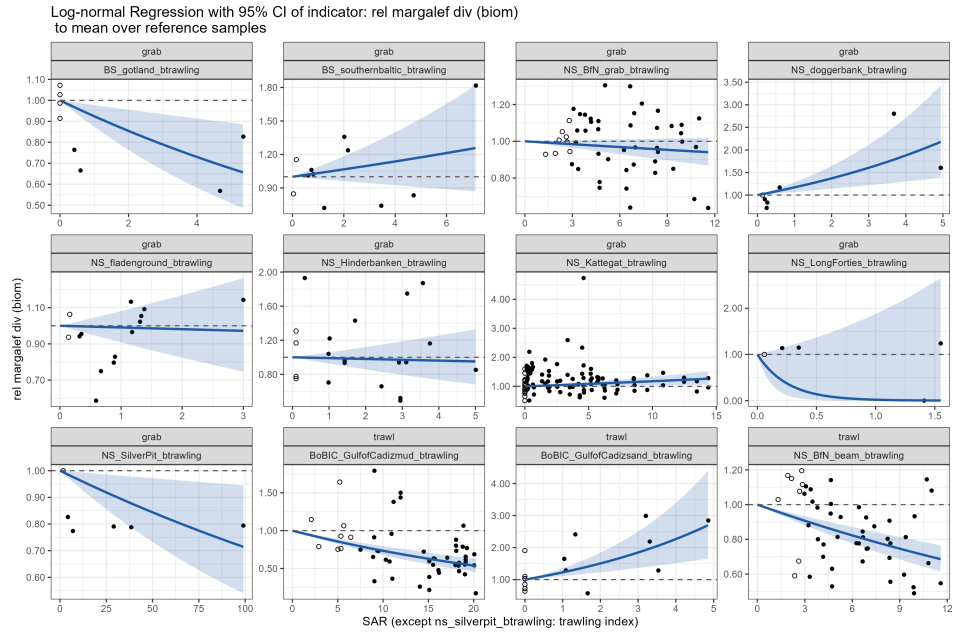


Figure 13: Pressure-state response curves for relative Margaleff diversity (biomass).